Computer Networks

1. Introduction to Computer Networks

A computer network connects multiple computers so they can exchange data and share resources. Networks are built from hosts, switches, and routers connected by links such as copper cable, fiber optics, or wireless radio.

The primary purpose of a network is reliable communication. Data is broken into small units called packets, which travel independently through the network and are reassembled at the destination. Packets carry the data itself together with headers that hold addressing and control information.

Networks can be described by their size and scope. A local area network, or LAN, spans a single building or campus and is usually owned by one organisation. A wide area network, or WAN, spans larger geographic distances and often connects several LANs together using leased lines or the public Internet.

Addressing is fundamental to communication. Every device on a network needs an address so that packets can be delivered to the correct destination. On the modern Internet this addressing is provided by the Internet Protocol, or IP, which assigns each interface a unique IP address.

2. The OSI and TCP/IP Models

The OSI reference model divides network communication into seven layers: physical, data link, network, transport, session, presentation, and application. Each layer provides services to the layer above it and hides the details of the layers below.

The physical layer deals with transmitting raw bits over a link. The data link layer moves frames between directly connected nodes and handles error detection. The network layer routes packets between hosts across multiple links, and the transport layer provides end-to-end delivery of data between applications.

The TCP/IP model is the practical model used on the Internet. It has four layers: link, internet, transport, and application. The internet layer handles addressing and routing, and the transport layer handles end-to-end delivery of data between applications.

Layering is a design principle that keeps each layer relatively simple and interchangeable. Because each layer exposes a well defined interface to the layer above, a change in one layer, such as replacing the physical medium, does not force a change in the other layers, which is why so many very different link technologies can all carry the same Internet traffic.

3. Routing and Latency

Routing is the process of choosing the path that packets take from source to destination. Routers exchange information about the network topology and use this to forward each packet toward its destination.

Each router keeps a forwarding table that maps destination addresses to the next hop on the path. When a packet arrives, the router looks up the destination in its table and forwards the packet to the next router. This process repeats at every router until the packet reaches its final destination.

The path a packet takes is typically not the shortest by distance but the path with the lowest overall cost, where cost can reflect bandwidth, delay, reliability, or administrative policy. Routing protocols allow routers to learn the topology and to adapt when links fail or new links become available.

Latency is the time it takes for a message to travel from sender to receiver. It includes transmission delay, propagation delay, and queueing delay at routers. Low latency is important for interactive applications such as video calls and online gaming.

Propagation delay is the time for a signal to cross a physical link, which grows with the distance the link covers. Transmission delay is the time to push a packet of a given size onto the link, which grows with packet size and shrinks as bandwidth grows. Queueing delay is the time a packet spends waiting in a router buffer, which grows as the network becomes congested.

Throughput is the amount of data that can be moved per unit of time, usually measured in bits per second. A link with high bandwidth has the capacity for high throughput, but real throughput is also limited by latency and congestion because the sender cannot push more data into the network than the slowest link and the queues along the path will accept.

4. Reliable Transport

The Transmission Control Protocol, or TCP, provides reliable, ordered delivery of a stream of bytes between two applications. It detects lost packets and retransmits them, and it controls the rate of sending to avoid overwhelming the network.

TCP establishes a connection before exchanging data. Each byte stream is broken into segments, and each segment carries a sequence number so that the receiver can detect losses and put out-of-order data back in the correct order. When the receiver acknowledges received data, the sender can infer which segments were lost.

Because the network can lose or reorder packets, TCP must cope with uncertainty. A timeout or a set of duplicate acknowledgements tells the sender that a segment was lost, and the sender retransmits it. This end-to-end reliability is what makes TCP the default choice for web browsing, email, and file transfers.

TCP also performs congestion control. It slowly increases its sending rate until a loss is detected, then backs off, so that many TCP flows sharing a link can split the available bandwidth fairly without overloading the network.

The User Datagram Protocol, or UDP, provides a lightweight, connectionless service. UDP adds no guarantees of delivery or ordering, but it has low overhead and low latency, making it useful for real-time applications such as voice and video where a delayed or retransmitted packet is not worth waiting for.

Because UDP does not establish a connection and does not track each segment, its header is much smaller than TCP's header and the protocol imposes no retransmission delay. This makes UDP attractive for DNS lookups, live streaming, online gaming, and voice over IP, where latest data matters more than reliable delivery of older data.